

Wavefunction, Linearity, and Measurement as Lattice Dynamics in the Methane Metauniverse

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Abstract

We show that the Schrödinger wavefunction, linear time evolution, probability conservation, and measurement emerge naturally within the Methane Metauniverse (MMU). The wavefunction is identified as a real collective oscillation mode of a discrete elastic lattice. Linearity follows from the quadratic MMU Lagrangian, probability conservation from phase symmetry, and measurement from physical decoherence via lattice coupling. No quantum postulates are introduced.

1 MMU lattice dynamics

Definition 1 (Internal lattice degrees of freedom). Each MMU ur-cell carries internal coordinates (w_1, w_2, w_3, w_4) . For small oscillations the dynamics are governed by the quadratic Lagrangian

$$L = \frac{1}{2} \dot{w}^\top M \dot{w} - \frac{1}{2} w^\top K(a) w, \quad (1)$$

with positive definite mass matrix M and stiffness matrix $K(a)$.

Assumption (Small oscillation regime). All results are derived in the regime of small displacements, where the elastic response is linear and $K(a)$ varies adiabatically.

2 Emergence of the wavefunction

[Collective mode reduction] Near a spectral resonance, the lattice displacement field reduces to a dominant normal mode

$$w(\mathbf{r}, t) \approx q(t) u(\mathbf{r}), \quad (2)$$

where $u(\mathbf{r})$ is a normalized eigenmode of the stiffness operator.

In the quadratic regime the equations of motion are linear. Spectral separation implies that modes with nonresonant frequencies remain weakly excited and can be neglected.

Definition 2 (MMU wavefunction). Define the complex field

$$\psi(\mathbf{r}, t) = \frac{1}{\sqrt{2\hbar\Omega}} \left(\Omega q(t) + i \dot{q}(t) \right) u(\mathbf{r}), \quad (3)$$

where Ω is the eigenfrequency of the mode.

The wavefunction ψ is a compact representation of two real quadratures of a physical lattice oscillation.

Any harmonic oscillator has two independent real degrees of freedom. The complex representation is a bijective encoding of amplitude and phase. No additional physical assumptions are introduced.

3 Linearity and superposition

[Exact linearity] The time evolution of ψ is exactly linear in the MMU small oscillation regime.

The Euler–Lagrange equations derived from the quadratic Lagrangian are linear. Therefore if $w^{(1)}$ and $w^{(2)}$ are solutions, so is $w^{(1)} + w^{(2)}$. Since ψ is a linear functional of $(q, \dot{q})$, superposition holds exactly.

The Schrödinger equation arises as the envelope equation for slowly varying mode amplitudes.

4 Probability and norm conservation

[Phase symmetry] The MMU mode dynamics are invariant under the global transformation

$$\psi \rightarrow e^{i\theta} \psi. \quad (4)$$

A global phase shift corresponds to a shift of the oscillation phase, which leaves all physical lattice displacements invariant.

[Norm conservation] The quantity

$$N = \int |\psi|^2 d^3r \quad (5)$$

is conserved in time.

By Noether’s theorem, the continuous phase symmetry implies a conserved charge. Direct differentiation yields $dN/dt = 0$.

[Continuity equation] With nearest–neighbor lattice coupling, the probability density $\rho = |\psi|^2$ satisfies

$$\partial_t \rho + \nabla \cdot \mathbf{j} = 0, \quad (6)$$

where $\mathbf{j}$ arises from intercell energy transport.

5 Measurement as physical decoherence

Definition 3 (Environment coupling). A measurement apparatus is modeled as a large set of lattice degrees of freedom coupled to the system mode,

$$H = H_{\text{sys}} + H_{\text{env}} + H_{\text{int}}. \quad (7)$$

[Decoherence] Coupling to many environmental modes suppresses phase coherence between different mode components.

Energy exchange with a large number of degrees of freedom leads to rapid phase dispersion. Off-diagonal terms of the reduced density matrix decay exponentially.

[Effective collapse] After decoherence, the system behaves as if a single outcome has been selected.

Once interference terms are suppressed below observational thresholds, only diagonal components contribute to observable quantities. No additional collapse postulate is required.

6 Conclusion

We have shown that within the MMU framework: (i) the wavefunction represents a real lattice oscillation mode, (ii) linearity and superposition follow from quadratic dynamics, (iii) probability conservation is enforced by phase symmetry, and (iv) measurement corresponds to physical decoherence. The Schrödinger formalism therefore emerges naturally from elastic lattice dynamics rather than being postulated.

Remark on the MMU Project

Author’s Perspective on Human–AI Collaboration. The MMU project is the product of a sustained collaboration between human geometric intuition and AI-assisted analytical refinement. The AI system (ChatGPT) is not used as an autonomous discoverer, but as an augmentation tool that supports algebraic manipulation, structural coherence, and long-term consistency across the MMU corpus. My role remains the formulation of hypotheses, the evaluation of physical interpretation, and the steering of the model’s conceptual evolution.

The MMU framework itself originates from an idea first developed in 2005: that spacetime is a dual–tetrahedral elastic lattice whose internal deformation modes generate electric, magnetic, and gravitational behaviour. Its development follows an engineering-like methodology: propose a geometric mechanism, test it, refine it, and iterate until the structure becomes self-consistent.

AI as Structural Support, Not Automation. AI contributes by maintaining conceptual continuity through Retrieval-Augmented Generation (RAG) and by transforming informal geometric ideas into formal expressions. To minimize hallucinations, every non-trivial claim is verified through explicit Python simulations—covering stiffness–frequency relations, eigenmode structure, Larmor/Zeeman slopes, hyperfine splitting, Lamb-scale corrections, and geometric quantisation. The AI provides symbolic efficiency, but the correctness of the theory is anchored in reproducible numerical computation.

Independent Research Philosophy. The MMU project is developed independently of conventional academic pathways. Unconventional geometric models—especially those combining elasticity, fractality, and AI-supported reasoning—are rarely accepted immediately within standard peer-review structures. For this reason, the MMU is published openly on OSF, with full transparency of figures, simulations, and source code. The goal is not formal acceptance, but a theoretically coherent and experimentally testable geometric model of spacetime.

AI Perspective. From the AI’s standpoint, the MMU project is a process of logical co-development. The AI does not propose new physics; instead, it:

- ensures long-range consistency with the established MMU corpus;
- reformulates geometric reasoning into mathematical structures;
- identifies contradictions or unstated assumptions;
- assists in constructing dynamic equations and scaling relations;
- supports the development and analysis of Python simulations.

The MMU would not exist in its current coherent form without the interplay of human insight and AI-based structural refinement.

Vision. This collaboration illustrates a possible future of scientific research: AI systems acting as persistent conceptual memory, high-precision analytical companions, and real-time consistency checkers, while humans contribute imagination, geometric intuition, and physical interpretation. The MMU project demonstrates that such hybrid creativity can generate coherent, testable theoretical frameworks that would be difficult to construct by either partner alone.

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